

PS 26043 / SMART EDUCATION / SOFTWARE

JANSETU

Community challenges to validated solutions.

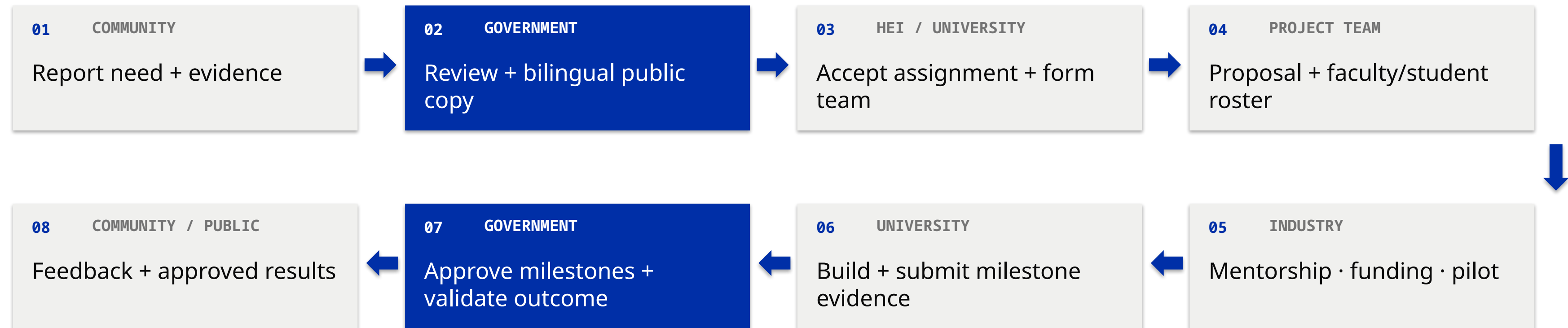
A bilingual platform to crowdsource local problems and coordinate Government, universities and industry through one accountable lifecycle.

Organization Government of Jharkhand
Department Higher & Technical Education
Team BilluSena
Team ID [Registered Team ID]

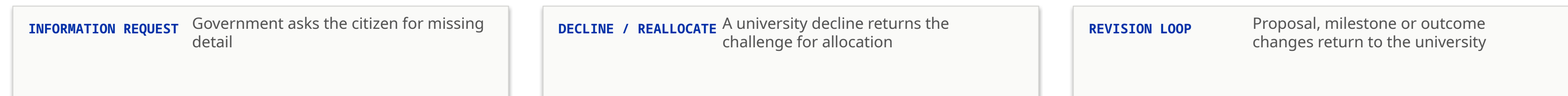
WORKING MVP · DEPLOYED



ONE PLATFORM · ONE ACCOUNTABLE LIFECYCLE



CONTROLLED BRANCHES



GEMINI ASSISTS

Classification / priority rationale / bilingual drafts / duplicate hints / university matches / outcome evidence gaps

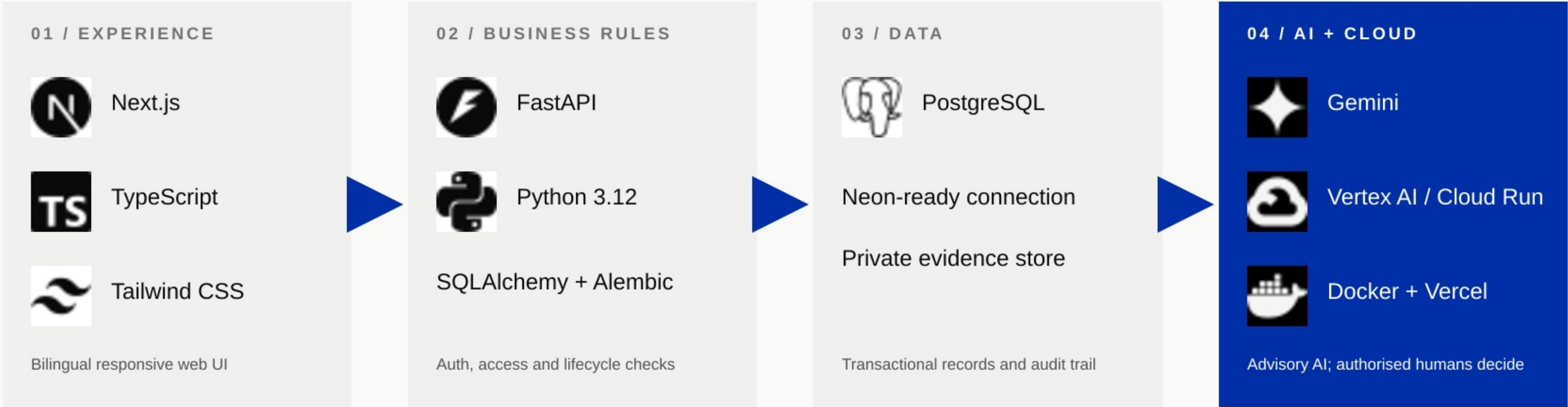
HUMANS DECIDE

Publish · assign · approve · validate

Browser /api/v1 → FastAPI business rules → PostgreSQL + private evidence ↔ Gemini advisory services

JANSETU / DEPLOYED MVP

Technology stack and service boundary



Browser /api/v1 → FastAPI → PostgreSQL · private storage ↔ Gemini advisory services

SECURITY

Argon2 · random DB sessions · HttpOnly cookies · origin checks · role + ownership rules

EVIDENCE

Private disk storage · signature checks · 5 files × 20 MB · authorised downloads

BUILDABLE NOW · EXPANDABLE WITH GOVERNMENT ADOPTION

01 / TECHNICAL

- Next.js, FastAPI and PostgreSQL are proven production technologies.
- Transactional lifecycle and access rules stay in the API.
- Structured AI responses are validated before display.

02 / ECONOMIC

- Open-source core with usage-based hosting.
- Neon, Cloud Run and Vercel scale independently.
- Funding commitments remain separate from money received.

03 / OPERATIONAL

- Four workspaces follow real stakeholder responsibilities.
- Human approval protects government accountability.
- Seed and migrations provide a repeatable demonstration.

RISKS → BUILT-IN RESPONSE

AI unavailable

Save first; retry transient failures; keep manual review.

Unsafe public copy

Minimum lengths + bilingual government approval.

Access / upload abuse

Role checks, origin validation and file signatures.

Ephemeral file storage

Move evidence to private object storage before rollout.

PROPOSED ROLLOUT

NOW

Deployed MVP

01

Pilot districts

02

Participating HEIs

03

Statewide network

Rollout phases are proposed; field adoption is not claimed.

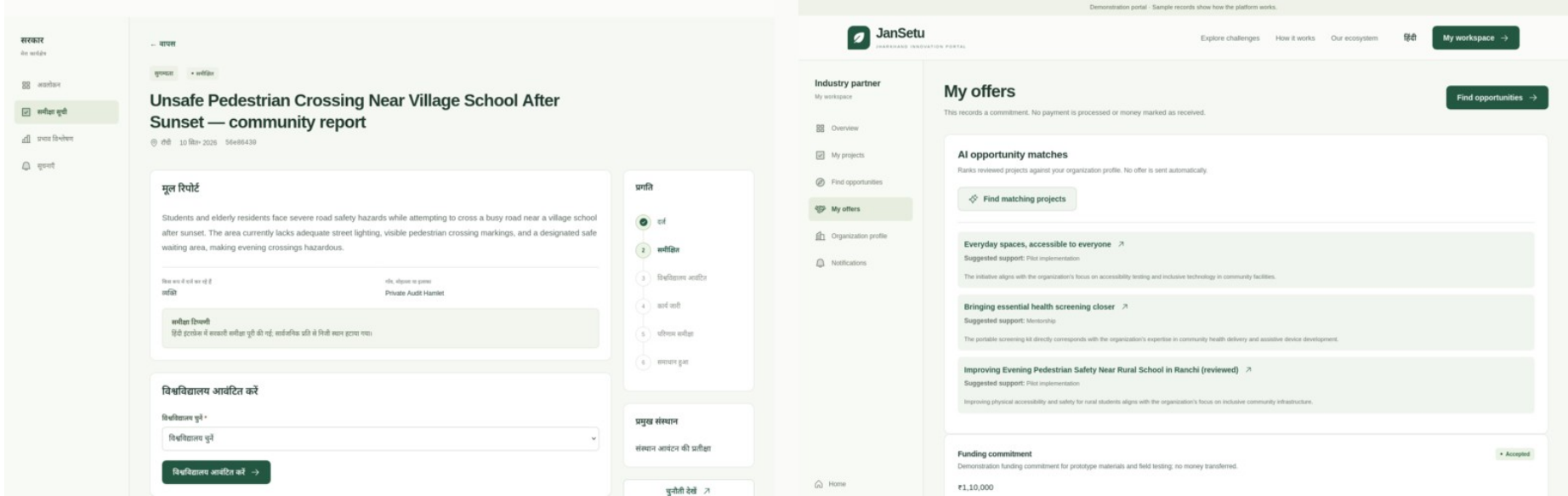
MEASUREMENT CHAIN



VALUE FOR EACH PARTICIPANT

01 / CITIZEN	Bilingual reporting, progress visibility and optional feedback.
02 / GOVERNMENT	Review, allocation, approvals, audit history and analytics.
03 / UNIVERSITY	Real problem statements, multidisciplinary teams and delivery.
04 / INDUSTRY	Reviewed opportunities matched to practical support.

WORKING MVP EVIDENCE



GOVERNMENT REVIEW · HINDI

INDUSTRY AI MATCHING

SEEDED DEMO SCALE

7	10	5	6
demo accounts	demo challenges	institutions	lifecycle stages

Public pages expose approved summaries and aggregate outcomes only. These numbers demonstrate MVP coverage, not measured field impact.

DEMO + SOURCE

FRONTEND

jansetu-jharkhand.vercel.app

API HEALTH

jansetu-api-595351864949.us-east4.run.app/api/v1/health

SOURCE CODE

github.com/kdxgautam/SIH-jansetu

PROBLEM CONTEXT

WHY JANSETU

- SIH 26043 asks for crowdsourced societal challenges and collaborative problem solving.
- NEP 2020 supports experiential, multidisciplinary and industry-linked learning.
- The portal connects those goals through an auditable government workflow.

PRIMARY TECHNICAL REFERENCES

AI

Gemini structured output · ai.google.dev/gemini-api/docs/structured-output

AUTH

Vertex ADC · cloud.google.com/vertex-ai/generative-ai/docs/start/quickstart

WEB

Next.js · nextjs.org/docs

API

FastAPI · fastapi.tiangolo.com

DATA

PostgreSQL · postgresql.org/docs / Neon · neon.tech/docs

CLOUD

Cloud Run · cloud.google.com/run/docs / Vercel · vercel.com/docs

DEMO STATUS

Seeded data · no field-impact claims · not an official government service